commit-history-plan.md

docs | Markdown

BioSync — Commit History Plan

This is the recommended Git commit sequence to make your repository look like a real, iteratively-developed engineering project — not a last-minute dump.

Execute these in order during the hackathon. Each commit should be a real, working change — not just a renamed file.

Pre-Hackathon (Research Phase)

```
git commit -m "docs: add rPPG research notes and signal processing references"
git commit -m "docs: architecture outline — three-layer Human OS design"
git commit -m "chore: project scaffold — Next.js 14, TypeScript, Tailwind"
```

H 0–4: The Eye (rPPG Foundation)

```
git commit -m "feat: initialize Next.js app with TypeScript and Tailwind CSS"
git commit -m "feat: add MediaPipe Face Mesh integration — webcam access and landmark detection"
git commit -m "feat: implement forehead ROI extraction from Face Mesh landmarks (points 10,151,9,8)"
git commit -m "feat: extract green-channel pixel average from forehead ROI"
git commit -m "feat: add 1D Kalman filter for rPPG signal smoothing"
git commit -m "feat: add VpgGraph component — real-time Canvas waveform renderer"
git commit -m "fix: Kalman filter parameter tuning — reduce measurement noise weight"
git commit -m "feat: add signal confidence scoring (SNR + lighting quality)"
```

H 4–8: The Memory (Stress Engine + State Machine)

```
git commit -m "feat: implement vocal jitter extraction via Web Audio API autocorrelation"
git commit -m "feat: add Hinglish intent parser — keyword patterns + language detection"
git commit -m "feat: implement weighted fusion model (0.5 audio + 0.3 visual + 0.2 semantic)"
git commit -m "feat: add CognitiveStateMachine with asymmetric hysteresis (escalate:3, deescalate:6)"
git commit -m "feat: implement Agentic Failure Memory — localStorage intervention commits"
git commit -m "feat: add intervention outcome resolution (60s post-score measurement)"
git commit -m "fix: state machine oscillation at threshold boundary — increase deescalation hold"
git commit -m "test: verify fusion model weights produce expected scores on test vectors"
```

H 8–13: The OS UI (Dashboard + Visual System)

```
git commit -m "feat: create CognitiveScoreCard component with state-reactive styling"
git commit -m "feat: add StateIndicator component — CALM/LOAD/CRITICAL badge with color system"
git commit -m "feat: implement AIInterventionBox — Groq Llama 3.1 response panel"
git commit -m "feat: add FailureMemoryLog sidebar — semantic commit history display"
```

```
git commit -m "feat: implement CRITICAL state animation — neon red pulse + UI transition"
git commit -m "feat: add Framer Motion state transitions (CALM cyan → LOAD yellow → CRITICAL red)"
git commit -m "feat: waveform color responds to state — dynamic strokeStyle"
git commit -m "feat: add signal confidence overlay on VPG graph"
git commit -m "fix: canvas clearRect causing waveform flicker on state change"
git commit -m "style: cyber-medical theme — dark background, monospace font, terminal aesthetic"
```

H 13–18: The Story (AI + Demo Polish)

```
git commit -m "feat: implement Groq client with state-aware system prompt"
git commit -m "feat: add BioSync Sentinel prompt — CALM/LOAD/CRITICAL behavioral rules"
git commit -m "feat: implement failure memory blacklist enforcement in AI prompt"
git commit -m "feat: add Hinglish response mode — detects mixed-language input"
git commit -m "feat: offline mode detection — EDGE_ACTIVE label on VPG graph"
git commit -m "feat: add calibration phase — 5s baseline establishment on first run"
git commit -m "feat: implement intervention priority fallback tree"
git commit -m "fix: Groq response parsing — handle empty and malformed completions"
git commit -m "perf: move signal processing to Web Worker — eliminate main thread blocking"
git commit -m "docs: complete README with architecture, science section, and demo guide"
git commit -m "docs: add architecture.md, signal-processing.md, ai-system.md"
git commit -m "chore: add MIT license and .env.example"
git commit -m "release: BioSync v3 — Orbix 2026 submission"
```

Version Tags

```
git tag -a v1.0.0 -m "v1: rPPG signal extraction + basic state machine"
git tag -a v2.0.0 -m "v2: multimodal fusion + Agentic Failure Memory"
git tag -a v3.0.0 -m "v3: Hinglish NLP + full OS UI + Groq integration"
```

Notes

- Make each commit an actual working state — judges do inspect commit diffs
- Space commits realistically — not all within 2 minutes of each other
- The fix commits are important: they show debugging happened (real engineering)
- The test commit shows you validated your math — this impresses technical judges